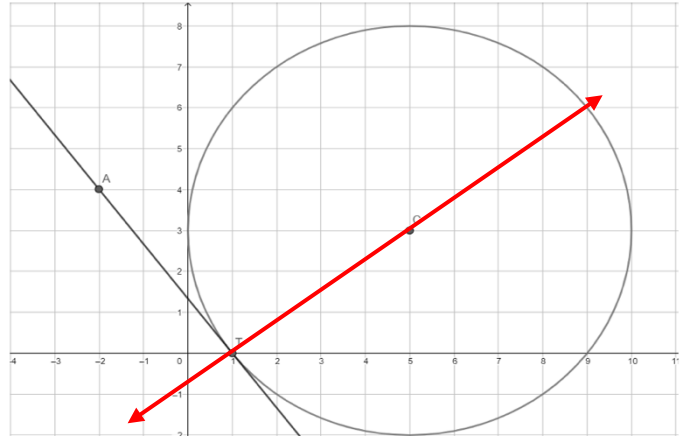


Geometry
Unit 13
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Use the following information for questions 1-4. Circle C is shown, where point T is on the circle and line AT is tangent to Circle C .



1. Use a straightedge to draw a line that goes through points T and C .

2. Write the equation of line AT .

$$y = -\frac{4}{3}x + \frac{4}{3}$$

3. Write the equation of line segment CT .

$$y = \frac{3}{4}x - \frac{3}{4}$$

4. Line AT and line CT intersect at point $(1, 0)$. Use the equations of line AT and CT to confirm this point.

$$\begin{aligned} y &= -\frac{4}{3}x + \frac{4}{3} \\ 0 &= -\frac{4}{3}(1) + \frac{4}{3} \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} y &= \frac{3}{4}x - \frac{3}{4} \\ 0 &= \frac{3}{4}(1) - \frac{3}{4} \\ 0 &= 0 \end{aligned}$$

Use the following information for questions 5-8. Circle N has a center at $(2, 3)$ and a tangent line with the equation $y = \frac{1}{2}x + 7$.

5. Write the equation of the line perpendicular to the tangent line.

$y = -2x + 7$

6. Find the point where the tangent line touches Circle N . Call this point D .

$\frac{1}{2}x + 7 = -2x + 7$ $y = -2(0) + 7$
 $x = 0$ $y = 7$
 $(0, 7)$

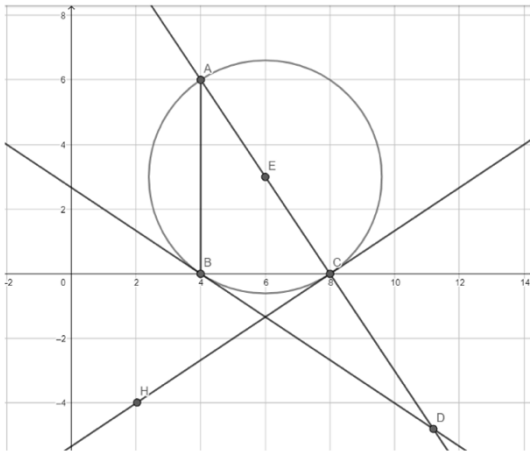
7. What is the radius of Circle N ?

$r = \sqrt{(2 - 0)^2 + (3 - 7)^2} = \sqrt{20} = 2\sqrt{5}$

8. Write the equation of Circle N .

$(x - 2)^2 + (y - 3)^2 = 20$

Use the following information for questions 9-10. Circle E is shown with secant line AD , tangent lines DB and CH , and chord AB .



9. Complete the table.

Line Segment	Length
$\overline{AB}$	6
$\overline{BC}$	4
$\overline{AC}$	$2\sqrt{13}$

10. Write the equation for circle E .

$(x - 6)^2 + (y - 3)^2 = 13$